Md Kawsher Mahbub

Department of Computer Science and Engineering
NPI University of Bangladesh, 495 East Basta, Singair, Manikganj-1800
Email: kawsher.cse@gmail.com Phone: +8801719145411 Web: kawsher.github.io
Google Scholar: scholar.google.com/citations?user=fQ_TTFsAAAAJ

RESEARCH INTERESTS

Artificial Intelligence, Deep Learning/Machine Learning, Medical Image Analysis, Health Informatics, Computer Vision, Uncertainty Quantification, Out-of-Distribution Generalization, Explainable AI

EDUCATION

Master of Science in Computer Science Aug. 2023 – May 2025
Kent State University, Kent, OH, USA
CGPA: 3.80

Bachelor of Science in Computer Science and Engineering Feb. 2017 – May 2021
Bangladesh University of Business and Technology (BUBT), Dhaka, Bangladesh
CGPA: 3.84

Higher Secondary Certificate (HSC) Jul. 2014 – Aug. 2016
Govt. Debendra College, Manikganj, Dhaka, Bangladesh
GPA: 4.67

Secondary School Certificate (SSC) Jan. 2009 – May 2014
Charkatari Sabuz Sena High School, Daulatpur, Manikganj, Dhaka, Bangladesh
GPA: 4.44

PROFESSIONAL EXPERIENCE

Assistant Professor Mar. 2026 – Present
Department of Computer Science and Engineering
NPI University of Bangladesh, Singair, Manikganj

Graduate Teaching Assistant Aug. 2023 – Dec. 2025
Department of Computer Science
Kent State University, Kent, OH, USA

Lecturer Dec. 2021 – Jul. 2023
Department of Computer Science and Engineering
Bangladesh University of Business and Technology (BUBT), Dhaka, Bangladesh

PUBLICATIONS

Under Review:

UR4. **Mahbub, M. K.**, & Biswas, M. (2026). STAMP: Predicting Out-of-Distribution Generalization without Target Data. arXiv preprint arXiv:2609.32672. Under review at ICLR 2027.

UR3. **Mahbub, M. K.**, Biswas M., Morshed M. N., and Yu. W. (2026). SpatialUQ: Post-Hoc Uncertainty Quantification from Spatial Consistency in Black-Box Vision Models. arXiv preprint arXiv:2610.09498. Under review at AISTATS 2027.

UR2. **Mahbub, M. K.**, Hossain, M. S., Miah, M. A. M., Morshed, M. N., Arpa, M. N., Modal, K. C., and Biswas, M. (2026). Beyond Benchmark Accuracy: Forensic Explainability Auditing for Ensemble Chest X-ray Classifiers, under review at Biomedical Signal Processing and Control, Elsevier.

UR1. **Mahbub, M. K.**, Hossain, S., Miah, M. A. M., Arpa, M. N., and Biswas M., (2026). Foundation Models in Healthcare: A Survey of Generalization, Uncertainty, and Trustworthiness, under review at ACM Computing Surveys, ACM.

Published Journal Papers:

J3. Ghose, P., Oliullah, K., **Mahbub, M. K.**, Biswas, M., Uddin, K. N., & Jamil, H. M. (2025). Explainable AI assisted heart disease diagnosis through effective feature engineering and stacked ensemble learning. Expert Systems with Applications, 265, 125928.

J2. **Mahbub, M. K.**, Biswas, M., Gaur, L., Alenezi, F., & Santosh, K. C. (2022). Deep features to detect pulmonary abnormalities in chest X-rays due to infectious diseases: COVID-19, pneumonia, and tuberculosis. Information Sciences, 592, 389-401.

J1. Ghose, P., Alavi, M., Tabassum, M., Ashraf Uddin, M., Biswas, M., **Mahbub, M. K.**, & Zhao, Z. (2022). Detecting COVID-19 infection status from chest X-ray and CT scan via single transfer learning-driven approach. Frontiers in Genetics, 13, 980338.

Published Conference Papers:

C7. **Mahbub, M. K.**, Zamil, M. Z. H., Miah, M. A. M., Ghose, P., Biswas, M., & Santosh, K. C. (2022, July). Mobapp4infectiousdisease: Classify COVID-19, pneumonia, and tuberculosis. In 2022 IEEE 35th International Symposium on Computer-Based Medical Systems (CBMS) (pp. 119-124). IEEE.

C6. **Mahbub, M. K.**, Biswas, M., Miah, M. A. M., & Kaiser, M. S. (2022, February). Deep neural networks for brain tumor detection from MRI images. In Proceedings of the Third International Conference on Trends in Computational and Cognitive Engineering: TCCE 2021 (pp. 473-485). Singapore: Springer Nature Singapore.

C5. **Mahbub, M. K.**, Miah, M. A. M., Islam, S. M. S., Sorna, S., Hossain, S., & Biswas, M. (2021, November). Best eleven forecast for Bangladesh cricket team with machine learning techniques. In 2021 5th International Conference on Electrical Engineering and Information Communication Technology (ICEEICT) (pp. 1-6). IEEE.

C4. **Mahbub, M. K.**, Biswas, M., Miah, A. M., Shahabaz, A., & Kaiser, M. S. (2021, July). COVID-19 detection using chest X-ray images with a RegNet structured deep learning model. In International Conference on Applied Intelligence and Informatics (pp. 358-370). Cham: Springer International Publishing.

C3. Biswas, M., **Mahbub, M. K.**, & Miah, M. A. M. (2021, December). An enhanced deep convolution neural network model to diagnose Alzheimer's disease using brain magnetic resonance imaging. In International Conference on Recent Trends in Image Processing and Pattern Recognition (pp. 42-52). Cham: Springer International Publishing.

C2. Biswas, M., Nova, A. J., **Mahbub, M. K.**, Chaki, S., Ahmed, S., & Islam, M. A. (2021, August). Stock market prediction: A survey and evaluation. In 2021 International Conference on Science & Contemporary Technologies (ICSCT) (pp. 1-6). IEEE.

C1. Biswas, M., Niamat Ullah Akhund, T. M., **Mahbub, M. K.**, Saiful Islam, S. M., Sorna, S., & Shamim Kaiser, M. (2021). A survey on predicting player's performance and team recommendation in game of cricket using machine learning. In Information and Communication Technology for Competitive Strategies (ICTCS 2020) ICT: Applications and Social Interfaces (pp. 223-230). Singapore: Springer Singapore.

SERVICES

Program Committee Member

- IEEE CBMS International Symposium on Computer-Based Medical Systems, 2023-Present
- The 35th International Conference on Artificial Neural Networks (ICANN 2026)

Reviewer for Journal Manuscript Submissions

- Journal of Imaging Informatics in Medicine (Springer)
- Expert Systems (Wiley)
- International Journal of Intelligent Transportation Systems Research (Springer)
- CMC - Computers, Materials & Continua (Tech Science Press)

- CMES - Computer Modeling in Engineering & Sciences (Tech Science Press)

TECHNICAL SKILLS

Programming Languages	Python, C, C++, C#, Java
Web Technologies	HTML5, CSS3, PHP, WordPress
Frameworks & Libraries	PyTorch, TensorFlow, OpenCV, NumPy, SciPy, Scikit-learn, Pandas, Keras, Matplotlib
Databases	MySQL, Oracle, MongoDB
Development Tools	Jupyter Notebook, Anaconda, PyCharm, Visual Studio, NetBeans, Eclipse, Android Studio
Miscellaneous	Git, LaTeX, UML, E-R Diagram

ACADEMIC PROJECTS

1. Crowd Source Recipe Collections (Master's Capstone Project)
2. Deep Neural Network for Pulmonary Abnormality Screening Using Chest X-ray: COVID-19, Pneumonia, and Tuberculosis
3. Infection Status from Chest X-ray and CT Scan via Single Transfer Learning Driven Approach
4. Hostel Management System

AWARDS AND HONORS

First Place and People's Choice Award, International Cook-Off 2024, Kent State University — Sep. 2024

Merit-based Scholarship (5 times), Bangladesh University of Business and Technology (BUBT)

LEADERSHIP

Head — Apr. 2026 – Present
ICT Cell at NPI University of Bangladesh

Secretary — Nov. 2024 – Sep. 2025
Bangladesh Student Association at Kent State University (BSA-Kent)

Advisor — Oct. 2022 – Jul. 2023
IEEE CS BUBT Student Branch Chapter

Member, Registration & Web Site Committee — Jul. 2022 – Oct. 2022
ICPC Asia Dhaka Regional Contest 2021

VOLUNTEERING / EXTRA-CURRICULAR ACTIVITIES

- Got 120+ accepted solutions at URI Online Judge (urionlinejudge.com.br · profile/193022)
- Got 50+ online judge accepted solutions (stopstalk.com/user/profile/kawsher11)
- Participated in intra-university programming contests

PROFESSIONAL MEMBERSHIP

Member, IEEE
Member, World Research Union